Asmitha Sathya

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EDUCATION

Johns Hopkins University - Baltimore, MD 21218

MSE Computer Science Graduation: May 2026

BS Computer Science, BS Biomedical Engineering | GPA: 3.70/4.00 Graduation: May 2025

Honors: Dean's List, Departmental Honors (CS + BME), General Honors, Linda Trinh Memorial Award

Relevant Coursework: Deep Learning, Explainable AI Design, Human-Robot Interaction, AI-ML for Global Health, Computer Vision, Intro to Robot Learning, Linear Signals & Systems, Systems & Controls, Build an Imager, Biomedical Data Science, Computer System Fundamentals

SKILLS & PROFICIENCIES

Languages & Libraries: C++, Python, Java, C, C#, R, MATLAB, JavaScript, TypeScript, XAML, Matplotlib, OpenCV, PyTorch, TensorFlow
Technologies: Git, Unity, React, Docker, Simulink, DDS, CMake, .NET, Linux/Unix, Arduino, Figma, Hugging Face, CUDA

WORK EXPERIENCE

Software Engineering Intern at Medtronic | *Unity, C#, MATLAB, Simulink, Docker, DDS, Git, Linux* *Jun 2025 - Aug 2025*
Medtronic, Boston, MA

- Built MATLAB/Simulink software integration testing simulation for Hugo RAS system, enabling hardware-free integration testing and reducing validation time by over 90%.
- Designed Unity VR simulation with movement tracking, logging of 9-DOF data, and playback functionality for formative user study.
- Collaborated with cross-functional team of 10 human factors, systems, hardware, and software engineers in fast-paced environment to iteratively update the VR simulation in parallel with evolving user needs and design requirements over 7 weeks.
- Developed mouse-controlled Hugo digital twin in Unity with accurate kinematics and physics movement for engineering purposes.

Software Engineering Intern at Triple Ring Technologies | *C#, XAML, Figma, .NET, GitHub* *Jun 2024 - Aug 2024*
Triple Ring Technologies, Newark, CA

- Utilized C# and XAML to develop GUI applications for 2 regulatory-compliant diagnostic medical devices intended for market release.
- Developed 100 black box and code inspection test cases in Helix ALM for traceability and acceleration of software verification by 50%.
- Enhanced Subversion to GitHub repository migration pipeline for non-standard directories, improving effectiveness by 10%.
- Nominated to attend Stanford Biodesign's rigorous 2024 DxD Healthtech Workshop, engaging with industry leaders.

Data Analytics Intern at National Institute of Standards and Technology | *R, STRmix, ggplot2, Plotly* *Aug 2023 - May 2024*
Dr. Sarah Riman, National Institute of Standards and Technology (NIST), Gaithersburg, MD

- Developed over 30 unique R scripts to curate, analyze, and visualize large-scale forensic DNA profiling data for publication.
- Programmed novel R pipeline that increased speed of labeling and differentiation of allelic and stutter peaks by 90%.
- Presented literature on challenges facing interpretation of casework DNA evidence to NIST experts for scope validation.

RESEARCH EXPERIENCE

Intelligent Medical Robotic Systems and Equipment Lab | *Python, LoRA, CUDA* *Aug 2025 - Present*
Dr. Axel Krieger, Laboratory for Computational Sensing + Robotics, Johns Hopkins University

- Designed pipeline to incorporate counterfactual learning via early experience into imitation learning paradigm to enhance generalization and accuracy of task and trajectory planning during autonomous surgery.
- Instruction fine-tuned VLM models with LIBERO dataset to produce natural language goals, yielding over 95% semantic similarity.
- Prototyping mean-preserving filtering (MPF) methods to denoise real X-ray projections for Gaussian Splatting-based CT reconstruction.

Advanced Robotics and Computationally Augmented Environments Lab | *Unity, C#, and Blender* *Sep 2023 - May 2024*
Dr. Mathias Unberath, Laboratory for Computational Sensing + Robotics, Johns Hopkins University

- Developed a programming interface using Unity, C#, and Blender to position a dynamic 3D model of the Loop-X mobile imaging robot along 6 coordinates.
- Reduced integration time of 3D digital twin Operating Room with virtual reality by 15%, enhancing surgical training.

Department of Biomedical Engineering Ahn Lab | *R, Python, TCGA, OpenCRAVAT, GDC, SigmaPlot* *Jul 2021 - May 2024*
Dr. Eun Hyun Ahn, Department of Biomedical Engineering, Johns Hopkins University

- Leveraged R and Python to create over 40 data visualizations and contributed to 2 publications on whole-exome and duplex sequencing of primary and recurrent glioblastoma tumors.
- Utilized GDC API, GeneCards API, and Python to create pipeline to annotate mutations with TCGA data and clinical significance.
- Produced 3 TCGA computational tutorials and taught to 150 students in biochemistry course (CTEI Instructional Enhancement Grant).

Bioinformatics and Computational Biology Lab | *Biopython, Snakemake, Bowtie 2, Linux* *Jun 2023 - Dec 2023*
Dr. Joel Bader, Department of Biomedical Engineering, Johns Hopkins University

- Employed Biopython and Snakemake in a Linux environment to analyze 177 barcode sequences generated by new multiplex detection of protein-protein interaction technology.
- Optimized mapping of merged and paired-end reads to reference genome, reducing processing time by 20%.

RELEVANT PROJECTS

WellSaid Patient Medication Portal | *React Native, Expo, TypeScript, Figma, OpenAI CLIP, Google Tesseract* *Fall 2025*
- Designed patient-friendly mobile application with automatic medication regimen creation and interactive pillbox setup voice assistant.
- Reduced pillbox setup errors by 60% compared to baseline in 20 participant user study, with a 95% average ease-of-use rating.

Multimodal Alerts in Surgical Task Management | *Python, Raspberry Pi, Pygame, Thonny, MicroPython* *Spring 2025*
- Programmed EKG-triggered alert system with 4 alert modalities, integrating hotspot socket communication to control robotic device.
- Conducted a user study with simulated surgical task, logging 1000+ data points from alert system to assess reaction time and accuracy.

Predicting Hemoglobin Levels for Anemia Severity Assessment | *Python, Pandas, NumPy, PyTorch, OpenCV, scikit-learn* *Fall 2024*
- Developed model for non-invasive anemia diagnosis via cellphone images of patients' palms, fingernails, tongue, and conjunctiva.
- Utilized YOLOv8 and EfficientNet for segmentation and regression in Python to determine hemoglobin level, achieving RMSE of 1.34.

Bone Age Prediction from X-Ray Images | *Python, OpenCV, SciPy, matplotlib* *Fall 2024*
- Utilized InceptionV3 with data augmentation to build a pediatric bone image classification model from 13000 X-ray images.
- Reduced MAE by 55.5% from previously accepted MobileNet model.

Query Ability of Probabilistic Data Structures | *Python, PyProbables, Sourmash API, Sequence Bloom Tree* *Fall 2024*
- Implemented bloom filter, quotient filter, and cuckoo filter manually with faster insertion time than PyProbables library functions.
- Developed efficient k-mer table storage method for large datasets using Sequence Bloom Tree and Sourmash API in Python.

LEADERSHIP

Computer Vision Course Assistant *Jan 2026 - Present*
- Delivered 30-minute lecture to 100 students on NumPy and image-processing fundamentals using Python in Google Colab.
- Held 2-hour weekly office hours to help students understand, implement, and debug classical and deep learning computer vision pipelines including image processing, feature detection, object recognition, and segmentation.
- Graded homework and quizzes for 100 students and delivered clear, actionable feedback to improve student understanding of concepts.

Johns Hopkins Design Team (Stomify) - 2024-25 Design Team Leader *Aug 2024 - May 2025*
Clinical Mentor: Dr. Andrew Cohen, MD, Urology, Johns Hopkins Medicine
- Led cross-functional team of 7 undergraduate students in developing a urostomy baseplate and pouch system for leakage management.
- Communicated technical concepts to non-technical stakeholders through 2 focus group sessions including 10 patients with urostomy.
- Dedicated 12 months to the design process to validate needs, map to design requirements, and conduct verification/validation.
- Awarded Linda Trinh Memorial Award for diligent efforts to improve the human condition and secured $1000 from Johns Hopkins University Summer Bridge Fund to support project continuation.

Computational Medicine: Cardiology Laboratory Teaching Assistant *Aug 2024 - Dec 2024*
- Instructed 120 undergraduate students in labs corresponding to analyzing electrocardiogram signals and the cardiovascular system.
- Dedicated 12 hours weekly to explaining complex material and providing instructional feedback on student reports.

JHU Zinda Dance Team - 2023-24 Co-Captain, 2022-23 Treasurer *Oct 2021 - May 2024*
- Choreographed 7 dance styles and led 19-member team in rehearsals and competitions (3rd in Intercollegiate Competition).
- Prepared budget proposals that secured $10,000 and raised $2,000 in monthly fundraisers to cover all expenses.

Johns Hopkins Design Team (LymphaLock) - Prototyping Lead *Aug 2022 - May 2023*
Clinical Mentors: Dr. Clifford Weiss, MD, Interventional Radiology; Dr. David Gage, MD, Radiology, Johns Hopkins Medicine
- Developed a device for interventional radiologists to maintain stable access to the lymph nodes during lymphangiographies.
- Conducted root cause, failure mode, and risk analysis to identify procedural issues and develop design requirements for solution.
- Employed 3D printing, CAD, laser cutting, soldering, and usability testing to design and build 2 functional works-like prototypes.
- Created 3 accurate lymph node polymer models based on extensive research, used for prototype verification and force testing.
- Recognized as Pava Center's Spark Accelerator Alumni, Hopkins New Venture Challenge Finalist, and 2023 Design Day Presenters.

PUBLICATIONS

Riman, S., Bright, J., Huffman, K., Moreno, L., Liu, S., **Sathya, A.**, & Vallone, P. (2024). A collaborative study on the precision of the Markov chain Monte Carlo algorithms used for DNA profile interpretation. *Forensic Science International: Genetics*. https://doi.org/10.1016/j.fsigen.2024.103088

Ruiz, A.L., Xiao, M., **Sathya, A.**, Piccininni, N., Liu, G., Siddiq, N., Chen, H., & McEnnis, K. (2023). Influence of particle z-potential and experimental procedure on protein corona formation and multicomponent aggregation. *AIChE Journal, 69*(12). https://doi.org/10.1002/aic.18237